

1 Scattering Applications

We are now equipped to calculate scattering cross sections. Recall from that the Born approximation gives the scattering amplitude as the Fourier transform of the potential evaluated at the momentum transfer $\vec{q} = \vec{k}' - \vec{k}$, with k the incoming momentum and k' the outgoing momentum:

$$f^{(1)}(\theta, \phi) = -\frac{2m}{4\pi} \tilde{V}(\vec{q}), \quad \tilde{V}(\vec{q}) = \int d^3x e^{-i\vec{q}\cdot\mathbf{x}} V(\mathbf{x}). \quad (1)$$

This is the leading term in the Born series: the incoming plane wave scatters only once off the potential and produces an outgoing spherical wave. Now we will apply this to a few important systems and potentials.

Scattering off a Yukawa potential

A massive mediator particle of mass m_{med} (such as the W or Z boson) gives rise to the Yukawa potential:

$$V_{\text{Yukawa}}(r) = \frac{\alpha_Y}{r} e^{-m_{\text{med}} r}, \quad (2)$$

where m_{med} has dimensions of inverse length in natural units and determines the range of the interaction. The Coulomb potential is recovered when $m_{\text{med}} = 0$ and $\alpha_Y = \alpha = e^2/(4\pi) = 1/137$.

Note that for $m_{\text{med}} \rightarrow \infty$, the Yukawa potential reduces to a contact interaction. Somewhat schematically, we can write

$$V_{\text{Yukawa}}(r) \simeq \frac{\alpha_Y}{r} e^{-m_{\text{med}} r} \xrightarrow{m_{\text{med}} \rightarrow \infty} \frac{4\pi\alpha_Y}{m_{\text{med}}^2} \delta^{(3)}(\mathbf{x}), \quad (3)$$

where we used the fact that $e^{-m_{\text{med}} r}/r$ approaches $4\pi\delta^{(3)}(\mathbf{x})/m_{\text{med}}^2$ in the limit of large m_{med} . This is the non-relativistic limit of the Fermi theory of weak interactions, where the W boson mass is much larger than the typical momentum transfer, leading to a contact interaction between fermions that is proportional to the Fermi constant $G_F \sim 1/m_W^2$. Other examples of Yukawa-like potentials include the nuclear potential mediated by pions and the screened Coulomb potential in a plasma, where the photon acquires an effective mass due to interactions with the medium.



The Yukawa potential arises from the Klein-Gordon equation in the presence of a point source of “charge” g at the origin:

$$(-\nabla^2 + m_{\text{med}}^2) \phi(\mathbf{r}) = g \delta^3(\mathbf{r}). \quad (4)$$

Away from the origin, spherical symmetry and the radial Laplacian reduce this to

$$-\frac{1}{r} \frac{d^2}{dr^2} (r \phi(r)) + m_{\text{med}}^2 \phi(r) = 0, \quad (5)$$

with general (spherically symmetric) solution

$$\phi(r) = \frac{A e^{-m_{\text{med}} r}}{r} + \frac{B e^{+m_{\text{med}} r}}{r}. \quad (6)$$

Demanding $\phi(r) \rightarrow 0$ as $r \rightarrow \infty$ kills the growing mode ($B = 0$), and matching the residual delta function at the origin fixes $A = g/4\pi$:

$$\phi(r) = \frac{g}{4\pi} \frac{e^{-m_{\text{med}} r}}{r}. \quad (7)$$

We can interpret $\phi(r)$ as the Green’s function of the Klein-Gordon operator, which describes the static field generated by a point charge at the origin. If we place a second particle of charge g' in the system, it experiences an interaction energy $V(r) = g' \phi(r)$, which is exactly the form of the Yukawa potential.

Scattering off a Yukawa potential

Let us compute the scattering amplitude in the Born approximation. Since V_{Yukawa} is central, we align the z -axis with $\vec{q}$ and perform the angular integration first:

$$\begin{aligned} \tilde{V}_{\text{Yukawa}}(\vec{q}) &= \alpha_Y \int d^3x \frac{e^{-i\vec{q} \cdot \mathbf{x}} e^{-m_{\text{med}} r}}{r} \\ &= 2\pi \alpha_Y \int_0^\infty dr r e^{-m_{\text{med}} r} \int_{-1}^1 d\cos\theta e^{-i|\vec{q}|r\cos\theta} = \frac{4\pi \alpha_Y}{|\vec{q}|^2 + m_{\text{med}}^2}. \end{aligned} \quad (8)$$

The momentum transfer for elastic scattering satisfies

$$|\vec{q}|^2 = |\vec{k}' - \vec{k}|^2 = 2k^2(1 - \cos\theta) = 4k^2 \sin^2(\theta/2), \quad (9)$$

where θ is the scattering angle and $k = |\vec{k}| = |\vec{k}'|$. Using $f^{(1)} = -(2m/4\pi) \tilde{V}(\vec{q})$ and $d\sigma/d\Omega = |f|^2$, we obtain the differential cross section:

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha_Y^2}{16 E^2} \frac{1}{\left(\sin^2(\theta/2) + \frac{m_{\text{med}}^2}{4k^2} \right)^2}}, \quad (10)$$

where $E = k^2/(2m)$ is the kinetic energy of the scattered particle. Another useful form is the differential cross section in terms of the momentum transfer $|\vec{q}|$ (using $d\Omega = 2\pi d\cos\theta = 2\pi d(1 - \cos\theta) = 2\pi d(|\vec{q}|^2/(2k^2))$):

$$\frac{d\sigma}{d|\vec{q}|^2} = 4\alpha_Y^2 \frac{m^2}{(|\vec{q}|^2 + m_{\text{med}}^2)^2}, \quad \text{with } |\vec{q}|_{\text{max}}^2 = (2k)^2, \quad |\vec{q}|_{\text{min}}^2 = 0. \quad (11)$$

Clearly, the largest momentum transfer corresponds to backwards scattering ($\theta = \pi$), when $\vec{k}' = -\vec{k}$.

Massless limit: Rutherford scattering. Setting $m_{\text{med}} = 0$ and taking $\alpha_Y = \alpha Z$ (for a nucleus of charge Z), we recover the Rutherford formula:

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 Z^2}{16 E^2 \sin^4(\theta/2)}. \quad (12)$$

The divergence as $\theta \rightarrow 0$ (forward scattering, $|\vec{q}| \rightarrow 0$) is a well-known consequence of the simplified assumptions of our scattering model. This implies that the total cross section is infinite, which is clearly unphysical. Part of the problem is that the idea of a cross section (as we defined it) for the Coulomb problem is technically ill-defined. The $1/r$ potential is long-range and does not admit asymptotic plane wave states. Any wave in our system is technically interacting with the potential at all times, including the incoming wave, which turns out not be a plane wave at all, it is distorted by the potential. That being said, the Rutherford formula is still an excellent approximation for scattering at sizable angles. If your beam has a given angular size on the detector plane, then requiring $\theta > \theta_{\text{min}}$ regulates the divergence. It was this large-angle result that Rutherford used to discover the compactness of the nucleus.¹



IR divergence: Looking at Eq. (11), we can already guess that the total cross section will diverge quadratically as $m_{\text{med}} \rightarrow 0$:

$$\sigma = \int d|\vec{q}|^2 \frac{d\sigma}{d|\vec{q}|^2} \sim 4\alpha_Y^2 m^2 \int \frac{d|\vec{q}|^2}{|\vec{q}|^4} \rightarrow 4\alpha_Y^2 m^2 \left(\frac{1}{|\vec{q}|_{\text{min}}^2} - \frac{1}{|\vec{q}|_{\text{max}}^2} \right). \quad (13)$$

This behavior at $|\vec{q}|_{\text{min}}^2 \rightarrow 0$ is a type of **infrared (IR) divergence**. They are a long-distance effects (as opposed to other divergences that can arise from large energy or short distances).

By not asking questions about the problematic regions of phase space, we can restrict our physical observables to those that are called infrared-safe observables (e.g., the cross section for scattering with a minimum deflection angle). Other kinds of IR divergences show up in strong interactions (QCD) and much of the theoretical progress in coming up with precision predictions for the LHC, for example, was developing IR-safe observables.

Heavy-mediator limit. In the opposite regime, $m_{\text{med}} \gg |\vec{q}|$:

$$\frac{d\sigma}{d\Omega} \simeq \frac{4m^2 \alpha_Y^2}{m_{\text{med}}^4}. \quad (14)$$

The cross section is isotropic and, in the non-relativistic regime, independent of the scattering energy. Note that you could have obtained the same result by applying Born's formula to the contact interaction in Eq. (3). At higher energies where relativistic kinematics apply, m^2 is replaced by E_{total}^2 and the cross section grows with energy. This is characteristic of processes mediated by heavy particles, such as weak interactions via the W and Z bosons. Of course, the growth cannot continue indefinitely: once $|\vec{q}| \sim m_{\text{med}}$, the full propagator structure becomes important and the cross section levels off.

What is the correction to this contact limit at low energies? We can expand the denominator in Eq. (10) for small k/m_{med} and find

$$\frac{d\sigma}{d\Omega} \simeq \frac{4m^2 \alpha_Y^2}{m_{\text{med}}^4} \left(1 - \frac{4k^2}{m_{\text{med}}^2} \sin^2(\theta/2) + \dots \right). \quad (15)$$

¹For more on Coulomb scattering, including the solutions to the full Coulombic scattering, see D. Tong's lecture notes <https://www.damtp.cam.ac.uk/user/tong/aqm/aqmtten.pdf>. Note that even including the full Coulombic amplitude, the total cross section is still infinite.

This makes sense: the corrections all vanish in the $\theta \rightarrow 0$ ($|\vec{q}| \rightarrow 0$) limit, where the contact approximation is exact. In fact, we also learn that the corrections get smaller as powers of velocity $v = k/m$, which is the natural expansion parameter for non-relativistic scattering (except $k^2/m_{\text{med}}^2 = v^2 m^2/m_{\text{med}}^2$ is the relevant parameter).



Electromagnetic processes favor small momentum transfers and therefore dominate at low energies. Weak processes, conversely, are mediated by heavy bosons whose propagators suppress rates at low $|\vec{q}|$. The weak force appears “weak” precisely because we ordinarily probe it at energies far below $m_W \simeq 80$ GeV; the large mass in the propagator denominators suppresses both cross sections and decay rates. At energies comparable to m_W and m_Z , the electromagnetic and weak interactions become comparable in strength, a hint of electroweak unification.

Partial wave content of Yukawa scattering. It is instructive to decompose the Born amplitude into partial waves. Using the expansion

$$f(\theta) = \sum_{\ell} (2\ell + 1) f_{\ell} P_{\ell}(\cos \theta), \quad (16)$$

from our scattering notes, we can project the Yukawa amplitude onto each ℓ by integrating against $P_{\ell}(\cos \theta)$. Note that the form above absorbs the phase shift factor $e^{i\delta_{\ell}} \sin \delta_{\ell}$ into the definition of f_{ℓ} . Writing $\mu = \cos \theta$, the Born amplitude becomes

$$f(\theta) = -\frac{m\alpha_Y}{k^2(\beta - \mu)}, \quad \beta \equiv 1 + \frac{m_{\text{med}}^2}{2k^2} > 1 \quad (17)$$

The projection onto partial wave ℓ gives

$$f_{\ell}^{(1)} = \frac{1}{2} \int_{-1}^1 d\mu f(\theta) P_{\ell}(\mu) = -\frac{m\alpha_Y}{2k^2} Q_{\ell}(\beta), \quad (18)$$

where $Q_{\ell}(\beta)$ is the Legendre function of the second kind. For $\ell = 0$ we have $Q_0(\beta) = \frac{1}{2} \ln(\frac{\beta+1}{\beta-1})$ (note that this definition is valid for $\beta > 1$, this is the external domain of Legendre functions, as opposed to the internal domain $|\beta| < 1$. Both diverge as $\beta \rightarrow 1$). The physics is encoded in Q_{ℓ} and we can ask what happens in the two limits of the Yukawa potential.

Heavy-mediator limit ($\beta \gg 1$): In this regime, we have

$$Q_{\ell}(\beta) \rightarrow \frac{(\ell!)^2 (2/\beta)^{\ell+1}}{(2\ell + 1)!}, \quad \text{with } \beta \simeq m_{\text{med}}^2/(2k^2), \quad (19)$$

so

$$\frac{f_{\ell}}{f_0} \sim \left(\frac{4k^2}{m_{\text{med}}^2} \right)^{\ell}. \quad (20)$$

Each successive partial wave is suppressed by $(k/m_{\text{med}})^2$, as expected from the expansion in Eq. (15). When $k \ll m_{\text{med}}$, only $\ell = 0$ (s-wave) contributes, and the scattering is isotropic. This is why the contact-interaction cross section we found in the heavy-mediator limit has no angular dependence.

Massless limit ($\beta \rightarrow 1^+$): $Q_{\ell}(\beta)$ diverges (logarithmically) for all ℓ , so infinitely many partial waves contribute. Let us return to our semiclassical interpretation. A particle with wavenumber k and impact parameter b carries angular momentum $\ell = kb$. The Yukawa potential has range $R \simeq 1/m_{\text{med}}$, so only particles with $b \lesssim R$ scatter appreciably, giving $\ell_{\text{max}} \sim kR = k/m_{\text{med}}$, which diverges as $m_{\text{med}} \rightarrow 0$. This is the physical origin of the divergence. Waves with infinitely large impact parameters can still scatter pff the potential.



Note how the angular structure of the Yukawa cross section interpolates smoothly from the isotropic contact limit to the forward-peaked Rutherford limit, controlled by the single parameter k/m_{med} .

- Contact ($k \ll m_{\text{med}}$), $\ell_{\text{max}} \rightarrow 0$ (s-wave) isotropic
- Intermediate ($k \sim m_{\text{med}}$), $\ell_{\text{max}} \sim 1$ few partial waves
- Coulomb ($k \gg m_{\text{med}}$), $\ell_{\text{max}} \gg 1$ strongly forward-peaked

Scattering off a collection of potentials

Let us now use the Born approximation for a collection of identical scatterers. Consider N identical scattering centers located at positions $\mathbf{r}_j$, each producing the same central potential $v(r)$. The total potential is

$$V(\mathbf{x}) = \sum_{j=1}^N v(|\mathbf{x} - \mathbf{r}_j|). \quad (21)$$

In the Born approximation, the scattering amplitude is proportional to the Fourier transform of V :

$$\begin{aligned} \tilde{V}(\vec{q}) &= \int d^3x e^{-i\vec{q}\cdot\mathbf{x}} \sum_{j=1}^N v(|\mathbf{x} - \mathbf{r}_j|) = \sum_{j=1}^N e^{-i\vec{q}\cdot\mathbf{r}_j} \int d^3y e^{-i\vec{q}\cdot\mathbf{y}} v(y) \\ &= \tilde{v}(\vec{q}) \sum_{j=1}^N e^{-i\vec{q}\cdot\mathbf{r}_j}, \end{aligned} \quad (22)$$

where we substituted $\mathbf{y} = \mathbf{x} - \mathbf{r}_j$ in each term and $\tilde{v}(\vec{q})$ is the Fourier transform of a single scatterer. Note that we are assuming each scatterer is identical so that we can pull $\tilde{v}(\vec{q})$ out of the sum. The total amplitude factorizes into a single-scatterer amplitude and what we call a form factor (or structure factor):

$$f_{\text{tot}}(\theta, \phi) = f(\theta, \phi) F(\vec{q}), \quad F(\vec{q}) \equiv \sum_{j=1}^N e^{-i\vec{q}\cdot\mathbf{r}_j}, \quad (23)$$

where $f = -(2m/4\pi) \tilde{v}(\vec{q})$ is the Born amplitude for a single center and $F(\vec{q})$ is the form factor. The differential cross section is then

$$\frac{d\sigma}{d\Omega} = |f|^2 |F(\vec{q})|^2 \quad (24)$$

Everything about the arrangement of the scatterers is encoded in $|F(\vec{q})|^2$. Let us ask some questions about the limiting cases of this form factor assuming the scattering sites are confined to a compact region of size $R \simeq Na$ (where a is the typical spacing between scatterers and N their number).

Coherent scattering: $\lambda \gg a$. When the de Broglie wavelength $\lambda = 2\pi/k$ of the incident particle is much larger than the typical spacing a between scatterers, the momentum transfer is small: $|\vec{q}| \lesssim 2k = 4\pi/\lambda \ll 2\pi/a$. In this regime, the phase $\vec{q} \cdot \mathbf{r}_j$ is small for all j , since $|\vec{q} \cdot \mathbf{r}_j| \lesssim |\vec{q}| \cdot Na \ll 2\pi$. Each exponential $e^{-i\vec{q}\cdot\mathbf{r}_j} \simeq 1$, so the form factor becomes

$$F(\vec{q}) \simeq N, \quad |F(\vec{q})|^2 \simeq N^2. \quad (25)$$

The scatterers contribute *coherently*: their amplitudes add in phase, and the cross section scales as N^2 :

$$\boxed{\frac{d\sigma_{\text{coh}}}{d\Omega} \simeq N^2 \frac{d\sigma_{\text{single}}}{d\Omega}}. \quad (26)$$

The incoming wave “sees” the collection as a single object with N times the scattering strength. This is the quantum analog of the fact that N antennas radiating in phase produce an intensity proportional to N^2 . Examples of this includes neutron and X-rays diffraction, neutrinos scattering off of nuclei (coherent neutrino-nucleus scattering, CEvNS, where each nucleon acts as a scatterer), and many others.

Incoherent scattering: $\lambda \ll a$. In the opposite limit, $\lambda \ll a$, the momentum transfer can be large enough that the phases $\vec{q} \cdot \mathbf{r}_j$ vary rapidly from one scatterer to the next. To compute the cross section, we expand

$$|F(\vec{q})|^2 = \sum_{j=1}^N \sum_{l=1}^N e^{-i\vec{q} \cdot (\mathbf{r}_j - \mathbf{r}_l)} = N + \sum_{j \neq l} e^{-i\vec{q} \cdot (\mathbf{r}_j - \mathbf{r}_l)}. \quad (27)$$

The diagonal terms ($j = l$) give N . For the off-diagonal terms ($j \neq l$), the phase $\vec{q} \cdot (\mathbf{r}_j - \mathbf{r}_l)$ oscillates rapidly as a function of $\vec{q}$ or as we average over the orientations of the target. These oscillating cross terms wash out, leaving only the diagonal contribution:

$$\boxed{\frac{d\sigma_{\text{incoh}}}{d\Omega} \simeq N \frac{d\sigma_{\text{single}}}{d\Omega}}. \quad (28)$$

The cross section now scales linearly with N : the *intensities* (not the amplitudes) add. Each scatterer contributes independently and we lose all interference information.



The coherent-to-incoherent transition is controlled by the product $|\vec{q}|a$, which compares the momentum transfer to the inverse spacing. When $|\vec{q}|a \ll 1$ (long wavelength), the probe cannot resolve the internal structure and the scattering is coherent ($\propto N^2$). When $|\vec{q}|a \gg 1$ (short wavelength), the probe resolves individual scatterers and the cross terms average away, giving incoherent scattering ($\propto N$). The transition between these regimes is governed by the form factor $|F(\vec{q})|^2$, which contains a wealth of information about the spatial arrangement of the scatterers.



Coherent forward scattering and the index of refraction. The coherent regime is most striking in the exact forward direction, $\theta = 0$, where $\vec{q} = 0$ and the form factor is exactly $|F(\vec{0})|^2 = N^2$ regardless of the arrangement of the scatterers. As we saw before, this means that a plane wave traversing a medium of number density n acquires a phase shift from the coherent superposition of all forward-scattered waves, giving rise to the index of refraction.



Consider N scatterers arranged in a one-dimensional chain with uniform spacing a , so that $\mathbf{r}_j = ja\hat{z}$ for $0 \leq j \leq N-1$.

1. Show that the form factor is

$$|F(\vec{q})|^2 = \frac{\sin^2(Nq_z a/2)}{\sin^2(q_z a/2)}, \quad (29)$$

where $q_z = \vec{q} \cdot \hat{z}$.

2. Verify that $|F|^2 \rightarrow N^2$ when $q_z a \rightarrow 0$ and that the average of $|F|^2$ over q_z gives N .
3. At what scattering angles do the sharp diffraction peaks (Bragg condition) appear?

Nuclear Form Factors

Take the a sphere of uniform charge density $\rho(r) = \rho_0 \Theta(R - r)$, where Θ is the Heaviside step function. We can use it as a model for the charge distribution of a nucleus, and compute the form factor for elastic scattering off this distribution. The Fourier transform of the potential is

$$\tilde{V}(\vec{q}) = \int d^3\mathbf{x} e^{-i\vec{q}\cdot\mathbf{x}} V(\mathbf{x}) = \int d^3x e^{-i\vec{q}\cdot\mathbf{x}} \int d^3x' \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \quad (30)$$

Interchanging the order of integration and using the convolution theorem, we can write this as

$$\tilde{V}(\vec{q}) = \tilde{\rho}(\vec{q}) \tilde{v}(\vec{q}), \quad (31)$$

where $\tilde{\rho}(\vec{q})$ is the Fourier transform of the charge density and $\tilde{v}(\vec{q})$ is the Fourier transform of the Coulomb potential $1/r$. The form factor is then

$$F(\vec{q}) = \frac{\tilde{V}(\vec{q})}{\tilde{v}(\vec{q})} = \tilde{\rho}(\vec{q}). \quad (32)$$

The Fourier transform of the uniform sphere is

$$F(\vec{q}) = \tilde{\rho}(\vec{q}) = 4\pi\rho_0 \int_0^R dr r^2 \frac{\sin(qr)}{qr} = 3 \frac{\sin(qR) - qR \cos(qR)}{(qR)^3}. \quad (33)$$

This is the well-known form factor for a uniform sphere. In nuclear physics, it goes by the name of the Helm form factor, and it is used to model the charge distribution of nuclei in electron scattering experiments. Note that our definition differs a little from above in the sense that $F(0) = 1$ here, while $F(0) = N$ in the previous section. The first zero of the form factor occurs at $qR \simeq 4.493$, which corresponds to the first “diffraction” minimum in the scattering cross section. In reality, the form factor ensures that the scattering process is relevant only when $qR \lesssim 1$.

Spin-dependent scattering

So far we have treated the projectile and target as spinless particles. Let us now add spin to this picture. The most general rotationally-invariant contact interaction between two spin- $\frac{1}{2}$ particles is

$$V(\mathbf{x}) = (a + b \vec{\sigma}_1 \cdot \vec{\sigma}_2) \delta^{(3)}(\mathbf{x}), \quad (34)$$

where $\vec{\sigma}_1$ and $\vec{\sigma}_2$ are the Pauli matrices acting on the projectile and target spins, and a, b are coupling constants with dimensions of inverse mass (in natural units). The term proportional to a is spin-independent, while the term proportional to b depends on the relative spin orientation.

Recall that $S_i = \frac{1}{2}\vec{\sigma}_i$ and that for two spin- $\frac{1}{2}$ particles,

$$\vec{\sigma}_1 \cdot \vec{\sigma}_2 = 4S_1 \cdot S_2 = 2(S^2 - S_1^2 - S_2^2) = 2\left(S(S+1) - 2 \times \frac{3}{4}\right) = \begin{cases} +1 & \text{triplet } (S=1), \\ -3 & \text{singlet } (S=0). \end{cases} \quad (35)$$

where $\hat{S}$ is the total spin. There are then two effective scattering potentials, one for each spin configuration:

$$V_t = (a + b) \delta^{(3)}(\mathbf{x}), \quad V_s = (a - 3b) \delta^{(3)}(\mathbf{x}). \quad (36)$$

Since the Fourier transform of $\delta^{(3)}(\mathbf{x})$ is simply 1, the Born amplitude for a contact interaction $V = C \delta^{(3)}(\mathbf{x})$ is isotropic:

$$f^{(1)} = -\frac{2\mu}{4\pi} C \quad (37)$$

where μ is the reduced mass of the two-body system. The scattering is pure s -wave, consistent with the heavy-mediator limit. The differential cross section $d\sigma/d\Omega = |f|^2$ then depends on which spin channel the system is in.

For an unpolarized beam and target, we average the cross section over initial spin states. The total spin of two spin- $\frac{1}{2}$ particles can be $S = 1$ (triplet, 3 states) or $S = 0$ (singlet, 1 state), giving statistical weights of $3/4$ and $1/4$ respectively:

$$\left\langle \frac{d\sigma}{d\Omega} \right\rangle_{\text{spin}} = \frac{3}{4}|f_t|^2 + \frac{1}{4}|f_s|^2 = \frac{\mu^2}{4\pi^2} \left[\frac{3}{4}(a+b)^2 + \frac{1}{4}(a-3b)^2 \right] = \frac{\mu^2}{4\pi^2} (a^2 + 3b^2). \quad (38)$$

Note that the cross-term ab cancels after spin averaging. The total unpolarized cross section is

$$\sigma_{\text{unpol}} = \frac{\mu^2}{\pi} (a^2 + 3b^2). \quad (39)$$

The factor of 3 in front of b^2 reflects the three triplet states that contribute to spin-dependent scattering. Note that there is no interference term (we did not calculate $|f_t + f_s|^2$) because the asymptotic states for these two channels are orthogonal.



Verify Eq. (38) by computing $\langle (\vec{\sigma}_1 \cdot \vec{\sigma}_2)^2 \rangle$ and $\langle \vec{\sigma}_1 \cdot \vec{\sigma}_2 \rangle$ for unpolarized spins, and using $\langle |a + b \vec{\sigma}_1 \cdot \vec{\sigma}_2|^2 \rangle = a^2 + 2ab\langle \vec{\sigma}_1 \cdot \vec{\sigma}_2 \rangle + b^2\langle (\vec{\sigma}_1 \cdot \vec{\sigma}_2)^2 \rangle$.

Coherent vs. incoherent scattering from multiple spins. We can now combine the spin structure with our earlier discussion of scattering off a collection of N targets. Consider N spin- $1/2$ particles at positions $\mathbf{r}_j$ with spins $\vec{\sigma}_j$. The total potential is

$$V(\mathbf{x}) = \sum_{j=1}^N (a + b \vec{\sigma}_{\text{proj}} \cdot \vec{\sigma}_j) \delta^{(3)}(\mathbf{x} - \mathbf{r}_j). \quad (40)$$

In the Born approximation, the scattering amplitude factorizes as before, but now each scatterer contributes a spin-dependent phase:

$$f = -\frac{\mu}{2\pi} \sum_{j=1}^N e^{-i\vec{q} \cdot \mathbf{r}_j} (a + b \vec{\sigma}_{\text{proj}} \cdot \vec{\sigma}_j). \quad (41)$$

We can separate this into spin-independent and spin-dependent parts:

$$f = -\frac{\mu}{2\pi} \left[a F(\vec{q}) + b \sum_{j=1}^N e^{-i\vec{q} \cdot \mathbf{r}_j} \vec{\sigma}_{\text{proj}} \cdot \vec{\sigma}_j \right], \quad (42)$$

where $F(\vec{q}) = \sum_j e^{-i\vec{q} \cdot \mathbf{r}_j}$ is the form factor from Eq. (23). The cross section involves $|f|^2$, which contains three types of terms.

Spin-independent part (a^2 terms): The form factor gives $|F(\vec{q})|^2$. In the coherent limit ($\lambda \gg a$), this contributes $\propto N^2$, exactly as before. The spin-independent interaction adds coherently because it does not depend on the spin state of any individual target.

Spin-dependent part (b^2 terms): These involve $\sum_{j,l} e^{-i\vec{q} \cdot (\mathbf{r}_j - \mathbf{r}_l)} \langle (\vec{\sigma}_{\text{proj}} \cdot \vec{\sigma}_j) (\vec{\sigma}_{\text{proj}} \cdot \vec{\sigma}_l) \rangle$. For the diagonal terms ($j = l$), we have

$$(\vec{\sigma}_{\text{proj}} \cdot \vec{\sigma}_j)^2 = 3 \quad (43)$$

(an operator identity for Pauli matrices), contributing $3Nb^2$. For the off-diagonal terms ($j \neq l$) with unpolarized, uncorrelated target spins, $\langle \vec{\sigma}_j \rangle = 0$, so these terms vanish. The spin-dependent part therefore scatters *incoherently*, contributing proportionally to N rather than N^2 .

Cross terms (*ab* terms): These involve $\langle \vec{\sigma}_{\text{proj}} \cdot \vec{\sigma}_j \rangle$ averaged over unpolarized target spins, which vanishes.

Combining these results, the unpolarized cross section for N targets in the coherent ($\lambda \gg a$) limit is

$$\sigma \simeq \frac{\mu^2}{\pi} (N^2 a^2 + 3Nb^2). \quad (44)$$

The spin-independent part enjoys the full N^2 coherent enhancement, while the spin-dependent part adds incoherently and only scales as N .

Other Applications (Non-Examinable)

Example: non-relativistic neutrino-electron scattering. A physically important application is the scattering of neutrinos on electrons. At low energies, the weak interaction is mediated by the heavy W and Z bosons ($m_W \simeq 80$ GeV, $m_Z \simeq 91$ GeV), so we are deep in the heavy-mediator regime and the interaction is well described by a contact potential. The non-relativistic reduction of the Standard Model electroweak Lagrangian gives an effective potential of the form Eq. (34):

$$V_{\nu e}(\mathbf{x}) = \frac{G_F}{\sqrt{2}} (c_V + c_A \vec{\sigma}_\nu \cdot \vec{\sigma}_e) \delta^{(3)}(\mathbf{x}), \quad (45)$$

where $G_F \sim 10^{-5} \text{ GeV}^{-2}$ is the Fermi constant and c_V, c_A are the vector and axial-vector couplings (vector and axial-vector come from the relativistic weak interactions, which for $v/c \ll 1$ reduce to the non-relativistic form above). Their values depend on the neutrino flavor:

Channel	c_V	c_A
$\nu_e - e$ (CC + NC)	$\frac{1}{2} + 2 \sin^2 \theta_W \simeq 0.96$	$\frac{1}{2}$
$\nu_\mu - e$ (NC only)	$-\frac{1}{2} + 2 \sin^2 \theta_W \simeq -0.04$	$-\frac{1}{2}$

where $\sin^2 \theta_W \simeq 0.231$ is the weak mixing angle, and we indicate whether the process receives contributions from charged-current (CC) W -exchange, neutral-current (NC) Z -exchange, or both.



For the particle folks: can you draw the diagrams? Why is there no CC contribution to $\nu_\mu - e$ scattering?

Identifying $a = G_F c_V / \sqrt{2}$ and $b = G_F c_A / \sqrt{2}$, the scattering lengths in the two spin channels are

$$f_t = -\frac{\mu}{\sqrt{2}\pi} G_F (c_V + c_A), \quad f_s = -\frac{\mu}{\sqrt{2}\pi} G_F (c_V - 3c_A), \quad (46)$$

where $\mu \simeq m_\nu$ is the reduced mass (since $m_\nu \ll m_e$). The unpolarized total cross section follows from Eq. (39):

$$\sigma_{\nu e} = \frac{m_\nu^2 G_F^2}{2\pi} (c_V^2 + 3c_A^2). \quad (47)$$

The factor of m_ν^2 makes this cross section extraordinarily small. The cosmic neutrino background ($T_\nu \simeq 1.95 \text{ K} \simeq 1.7 \times 10^{-4} \text{ eV}$) consists of neutrinos that are non-relativistic today (for $m_\nu \gtrsim T_\nu$), and their scattering cross section on electrons is suppressed by $(m_\nu/m_W)^4$. This is why the cosmic neutrino background has never been detected (and probably will not be anytime too soon). In contrast, relativistic neutrinos (such as those from reactors or the Sun) have cross sections that scale as $\sigma \sim G_F^2 E_\nu^2$, where the neutrino energy E_ν replaces the mass. The transition between these regimes is set by whether the neutrino is relativistic ($E_\nu \gg m_\nu$) or non-relativistic ($E_\nu \simeq m_\nu$).



Coherent elastic neutrino-nucleus scattering (CE ν NS). This result explains why coherent neutrino-nucleus scattering is dominated by the vector (spin-independent) coupling. A nucleus with Z protons and $\mathcal{N}$ neutrons has a coherent cross section proportional to $(Zc_V^p + \mathcal{N}c_V^n)^2$, where $c_V^p = \frac{1}{2} - 2\sin^2\theta_W \simeq 0.04$ and $c_V^n = -\frac{1}{2}$. Since $c_V^p \ll c_V^n$, the proton contribution is negligible and

$$\sigma_{\text{CENS}} \propto \mathcal{N}^2 (c_V^n)^2 \simeq \frac{\mathcal{N}^2}{4}. \quad (48)$$

The $\mathcal{N}^2$ enhancement makes this the dominant neutrino interaction channel at low energies ($E_\nu \lesssim 50$ MeV). It was first predicted by Freedman in 1974 and observed by the COHERENT experiment in 2017.

The axial (spin-dependent) contribution, by contrast, only scales as N_{unpaired} , the number of unpaired nucleon spins. For even-even nuclei (where all nucleon spins are paired), the axial contribution vanishes entirely, and the scattering is purely coherent.

Dark matter and electromagnetic form factors

If dark matter consists of weakly interacting massive particles (WIMPs) with mass m_χ , they scatter off nuclei in a detector through some interaction potential, and the Born approximation tells us how to calculate it. Our formalism is applicable to this scenario because WIMPs are non-relativistic ($v \sim 10^{-3}c$). The problem is we do not know what their interactions with nuclei look like, but we can classify them. Even though dark matter is electrically neutral, it is instructive to consider all the possible electromagnetic-like interactions it could have, spanning simple electric-charge-like couplings (which require the DM charge to be tiny) to higher multipole moments (which can be nonzero even for neutral particles). Let us compare these possibilities.

Millicharge. If dark matter carries a small electric charge $Q = \epsilon e$ ($\epsilon \ll 1$), it scatters off a nucleus of charge Ze via the Coulomb potential:

$$V_{\text{charge}}(r) = \frac{\epsilon\alpha Z}{r}. \quad (49)$$

This is the Yukawa/Rutherford problem we already solved. The Fourier transform gives

$$\frac{d\sigma}{dq^2} \propto \frac{\epsilon^2\alpha^2 Z^2}{q^4} |F(\vec{q})|^2, \quad (50)$$

The $1/q^4$ behavior means the cross section is strongly peaked at small momentum transfers (forward scattering), a direct consequence of the $1/r$ potential. This interaction is said to be *spin-independent*: it couples to the nuclear charge, allows for a full Z^2 coherent enhancement, and cares mostly about nucleon (in this case, proton) number rather other nuclear properties. Note that the form factor $|F(\vec{q})|^2$ encodes the finite size of the nucleus and gives $|F(\vec{0})|^2 = 1$ in the forward limit (this differs from the definition above by a factor of $1/Z$ in $F(\vec{q})$ and makes the coherent scaling more explicit).

Electric dipole moment. A spin- $\frac{1}{2}$ dark matter particle can have a permanent electric dipole moment d_χ . This couples the DM spin $\vec{\sigma}_\chi$ to the nuclear electric field $\vec{E} = Z\alpha \hat{r}/r^2$ via $V = -d_\chi \vec{\sigma}_\chi \cdot \vec{E}$, giving

$$V_{\text{EDM}}(\mathbf{x}) = -d_\chi Z\alpha \frac{\vec{\sigma}_\chi \cdot \hat{r}}{r^2}. \quad (51)$$

Since $\hat{r}/r^2 = -\vec{\nabla}(1/r)$, the Fourier transform brings down one power of $\vec{q}$ relative to the Coulomb case:

$$\tilde{V}_{\text{EDM}}(\vec{q}) = \frac{4\pi i d_\chi Z\alpha}{q^2} \vec{\sigma}_\chi \cdot \vec{q}. \quad (52)$$

The amplitude is proportional to $\vec{\sigma}_\chi \cdot \hat{q}/q$ rather than $1/q^2$, and squaring gives

$$\frac{d\sigma}{dq^2} \propto \frac{d_\chi^2 \alpha^2 Z^2}{q^2}. \quad (53)$$

Two features distinguish this from the charge case: (i) the $1/q^2$ falloff is much softer than the $1/q^4$ of Rutherford scattering, meaning the angular distribution is less forward-peaked (it is “only” logarithmically divergent now); (ii) the interaction depends on $\vec{\sigma}_\chi$, but it couples to the nuclear charge (Z^2), not the nuclear spin.

Magnetic dipole moment. A magnetic dipole moment μ_χ couples through $\vec{\sigma}_\chi \cdot \vec{B}$. Since a static nucleus produces no magnetic field, the leading interaction arises from the motional field $\vec{B}_{\text{eff}} = -\vec{v} \times \vec{E}$ seen by the moving DM particle in the nuclear Coulomb field. The effective potential in momentum space is

$$\tilde{V}_{\text{mag}}(\vec{q}) \propto \frac{\mu_\chi Z \alpha}{q^2} \vec{\sigma}_\chi \cdot (\vec{q} \times \vec{v}), \quad (54)$$

which gives

$$\left. \frac{d\sigma}{dq^2} \right|_{\text{mag}} \propto \frac{\mu_\chi^2 \alpha^2 Z^2 v^2}{q^2}. \quad (55)$$

The q -dependence is the same as the electric dipole case ($1/q^2$), but there is an additional factor of v^2 from the velocity-dependent coupling. Since dark matter in the Milky Way moves at $v \sim 10^{-3}c$, this velocity suppression makes the magnetic dipole cross section much smaller than the electric dipole cross section for similar coupling strengths.

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This momentum-transfer dependence has direct experimental consequences. Dark matter direct detection experiments measure the nuclear recoil energy $E_R = q^2/(2m_N)$, so $d\sigma/dE_R \propto d\sigma/dq^2$. A millicharged particle produces a recoil spectrum $\propto 1/E_R^2$ (steeply falling, most events at low energy), while a dipole interaction gives $\propto 1/E_R$ (flatter spectrum, more events at higher recoil energies). This difference in the spectral shape allows experiments to distinguish between different types of dark matter interactions, even before identifying the underlying particle physics model.

Dark matter annihilation

The partial wave decomposition is also useful to understand dark matter annihilation. While scattering involves a DM particle bouncing off a target, annihilation involves a DM particle and its antiparticle (or two identical DM particles) meeting and converting their rest mass into lighter particles: $\chi\bar{\chi} \rightarrow f\bar{f}$, for some final state particles f . This process determines the cosmological abundance of dark matter (the relic density) and is the target of indirect detection experiments searching for DM annihilation products in the galaxy.

Why partial waves matter for annihilation. For annihilation through a short-range interaction (contact or heavy-mediator), only the wavefunction at zero separation matters: the two particles must be at the same point to annihilate. Going back to our partial wave decomposition, the radial wavefunction for partial wave ℓ behaves as

$$R_\ell(r) \xrightarrow{r \rightarrow 0} \frac{(kr)^\ell}{(2\ell + 1)!!}, \quad (56)$$

where $k = \mu v$ is the relative momentum and μ is the reduced mass. The probability density at the origin is therefore

$$|R_\ell(0)|^2 \propto k^{2\ell} \propto v^{2\ell}. \quad (57)$$

This is the key result: the annihilation rate in partial wave ℓ scales as $v^{2\ell}$ at low velocities, because the *centrifugal barrier* $\ell(\ell+1)/r^2$ suppresses the wavefunction at the origin for higher partial waves.

The annihilation cross section times relative velocity is usually more convenient in this field, and can be conventionally expanded as

$$\boxed{\sigma v = a + b v^2 + \mathcal{O}(v^4)}, \quad (58)$$

where the constant a is the s-wave ($\ell = 0$) contribution and $b v^2$ is the p-wave ($\ell = 1$) contribution.



The $v^{2\ell}$ scaling has a simple semiclassical explanation. In the classical limit, a particle with impact parameter b and momentum k has angular momentum $\ell = kb$. To annihilate, the two particles must reach $r = 0$, but the centrifugal potential $\ell(\ell+1)/(2\mu r^2)$ creates a barrier of height $E_{\text{barrier}} \sim \ell^2/(2\mu R^2)$ at the interaction range R . For the particles to penetrate this barrier, they need kinetic energy $E \sim k^2/(2\mu)$ comparable to E_{barrier} , which requires $k \gtrsim \ell/R$. At low velocities ($k \rightarrow 0$), the tunneling probability through the barrier scales as $\sim (kR)^{2\ell} \sim v^{2\ell}$, matching the quantum result.